Polar Tuberculoid Leprosy

In **polar tuberculoid (TT) leprosy**, cell-mediated immunity is strong, as evidenced by spontaneous cure and the absence of downgrading to a less resistant form. The primary skin lesion is a well-defined **plaque**, often **annular** in configuration due to peripheral expansion and central clearing (Fig. 186-1). The border(s) of the annulus are sharply marginated. The lesion is typically **firmly indurated, elevated, erythematous, scaly, dry, hairless, and hypopigmented**, though clinical variation is common.

A nearby sensory nerve may be enlarged, but the lesion itself is characteristically **hypesthetic** and **anhidrotic**, with a maximum diameter of **10 cm**. Skin lesions are often **solitary**, especially in patients who are TT *de novo*. In contrast, those who upgrade to TT from borderline tuberculoid (BT) may have multiple lesions—usually no more than three. Despite sufficient immunity for potential self-cure, antibiotic therapy is recommended.

Histopathology

- **De novo TT lesions**: Small tubercles with prominent lymphocytic mantles.
- **Upgrading lesions** (from BT to TT): Abundant **Langhans giant cells**, **exocytosis into the epidermis**, and lymphocytic mantles.

Borderline Tuberculoid Leprosy

In **borderline tuberculoid (BT) leprosy**, immunologic resistance is strong enough to limit disease extent and slow bacillary replication, but insufficient for spontaneous cure. These patients are immunologically **unstable**—they may **upgrade** to TT or **downgrade** to borderline lepromatous (BL).

Clinical Features

Primary lesions include **plaques** and **papules**. Like TT, annular configurations are common with sharply marginated borders. **Satellite papules** around the main plaque are characteristic (Fig. 186-2). In darkly pigmented individuals, **hypopigmentation** may be prominent.

Compared to TT: - Less scaling - Less erythema - Less induration and elevation - Lesions may become **much larger** (>10 cm), sometimes involving an entire extremity over time

Peripheral Nerve Changes in Leprosy

Five common types of peripheral nerve abnormalities occur in leprosy:

1. **Nerve enlargement** – Often asymmetric; commonly affects superficial nerves such as the **great auricular, ulnar, radial cutaneous, superficial peroneal, sural**, and **posterior tibial nerves**.
2. **Sensory impairment** – Present within skin lesions.
3. **Nerve trunk palsies** – May occur with or without signs of inflammation; **silent neuropathy** can lead to sensory and motor deficits (weakness, atrophy), and chronic cases may develop **contractures**.
4. **Stocking-glove pattern of sensory impairment (S-GPSI)** – Reflects progressive **C-fiber degeneration**, starting with loss of **heat and cold discrimination**, followed by pain and light touch. Begins in acral areas and extends proximally, often sparing palms initially.
5. **Anhidrosis** – On palms or soles, indicating **sympathetic nerve involvement**.

Bacillary Load and Ridley's Classification

- In **TT** and most **BT** cases, **acid-fast bacilli are not detectable**.
- In **BB, BL, LLs**, and **LLp**, bacilli are **readily demonstrable**.

- **Ridley's immunological classification** is valuable for categorizing patients based on clinical, histological, and immunological features, particularly in assessing **host immunity**.